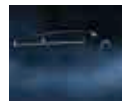




Hyundai buys Boston Dynamics

Hyundai Motor Group acquired an 80% stake in the company for 1.1 Billion dollars.
<https://digit.in/jul21-74>



Dodge to also go electric

Dodge will release an all-electric muscle car in 2024.
<https://digit.in/jul21-75>



The best AI musician

Which computer sounds the best?

Aditya Madanapalle | aditya@digit.in

M

ost music is inherently algorithmic, and is deeply linked to mathematics. Computational

approaches are a natural fit for music production, and the iterative nature of current AI technologies allows artists to interface with and guide the networks through the generation process. There are essentially three components of a song, the lyrics, the vocalisation and the instrumentation, although AI has been used for everything from generating the name of the bands, creating the album artwork and even making the accompanying music videos.

LYRICS

The simplest way for anyone to create the lyrics for a track is to use <https://theselyricsdonotexist.com> that straight up lets you choose a topic, a mood, a genre and lets you instantly generate suitable lyrics. A slightly more technical approach is a text based model such as GPT-3 for generating human like text based on a corpus that it has been trained on. Train it exclusively on traditional Japanese haiku, and the lyrics will be on those lines. A model is a program that has been trained to recognise particular patterns and replicate them on demand. There are models for all parts of making a song, from the lyrics, to the music, and the videos. The models for writing the lyrics can generate content based on a theme, a topic, or a starting "seed" line, or a corpus of text (which can be the lyrics of an existing band). It is at this point

that there is plenty of curation and editing. A lot of the time the text generated by the algorithms can be uninspiring or even unintelligible. Just sometimes, lightning strikes and the lyrics are just perfect. After this, there is a process of editing, which can involve reshuffling the sentences, fixing the lines or making adjustments. Some artists may only pick out the lyrics instead of editing them, to provide the AI with a greater share of the collaboration.

The pipeline for producing music in collaboration with AI has so many steps that the humans get to decide what part of the process they really want to work at. Some creators for example may choose to work with the training data, and curate just the perfect mix of tracks. Others may go for volume of production, throw computing power at the AI, and then be extremely picky about just choosing the best ones.